

Score: / 10

Calculus I (MATH 2200)

Spring 2020

Friday, 01/31/2020

Quiz 1: Section 2.1, 2.2

Name (Print): _____

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Section 2.1

2 pts

1. Suppose you throw a rock straight upwards with a speed of $96 \frac{ft}{s}$. Neglecting air resistance, a well-known formula from physics states that the position of the rock after t seconds is given by the function

$$s(t) = -16t^2 + 96t.$$

What was the *average velocity* of the rock between $t = 1$ s and $t = 4$ s?

2 pts

2. Let $f(x) = x^2 - 5$. Find the slope of the secant line between $(1, f(1))$ and $(5, f(5))$.

Section 2.2

3 pts 1. If $f(x) = \begin{cases} x+2 & \text{if } x \leq 0 \\ 2-x & \text{if } x > 0 \end{cases}$, find $\lim_{x \rightarrow 0} f(x)$.

3 pts 2. Find $\lim_{x \rightarrow 3} \frac{x-3}{x^2-9}$ or explain why the limit does not exist.